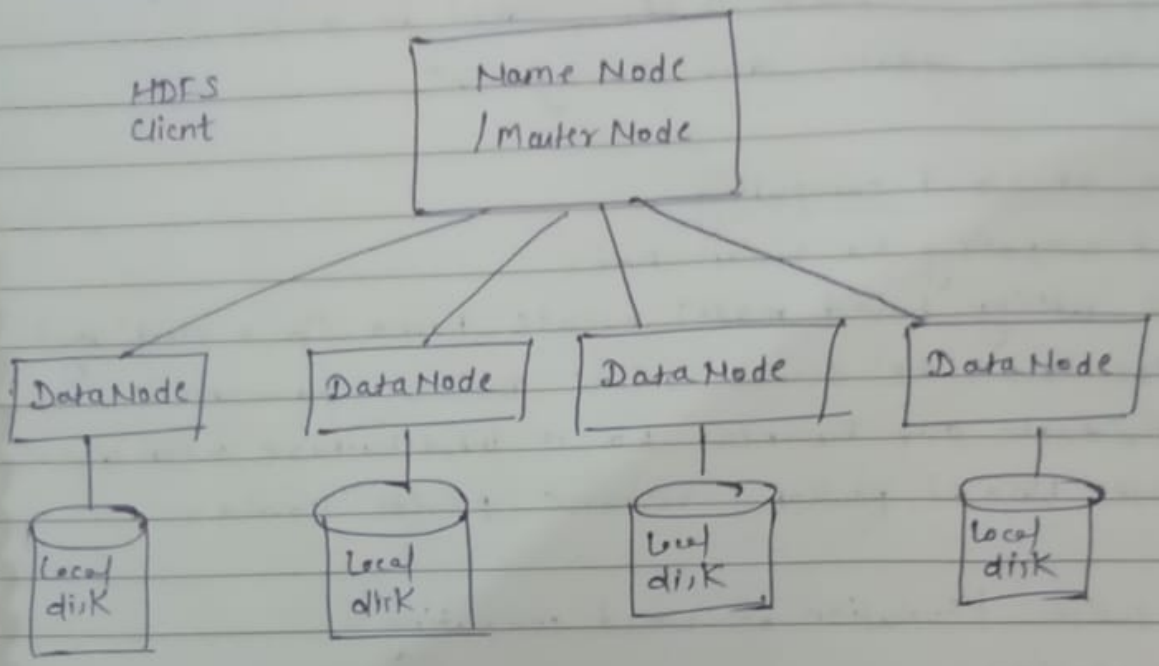
Date

Hadoop file System (HDFS)



HDFS → used for storing Big data / Huge number of data storing data in dividing part, Robust distributed file system.

Introduction to cloud computing.

definition:- Cloud computing ek aisa model hai jahan pe computing resources jaise ki server, storage, database, networking, software etc. Internet ke zariye access kiye jaate hai. User ko sirf 'Pay as you go' model main charge kiya jaata hai - matlab jitna use karo, utna hi paisa do.

Why cloud:-

Reason

No need for heavy Infrastructure

Scalability:- jitna chahiye utna badhao ya ghatao

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Cost efficient - hardware ya maintence ka tension nahi
Easy access :- kahin se bhi, kabhi be access.
Automatic updates aur management.

Vision of cloud computing :
Cloud computing ka vision hai -
- EK utility based model create karna (jaise electricity)
→ Har tarah ke services online available karwana.
→ Individuals aur business ko IT infrastructure mein invest kiye bina hi powerful resource provide karna.

Faundation / Evolution of cloud computing :-

Foundation

- Pehle ke computing models : Mainframe → Client server → Distributed system → Virtualization.
- Cloud computing evolved from :
  - Grid computing (resource sharing)
  - Utility computing (pay per use)
  - Virtualization (multiple OS on single hardware)

Evolution

1960s : John McCarthy ne bola tha computing ek day utility banegi
2000s : Amazon ne AWS launch kiya
Aaj : Google Cloud, microsoft azure, IBM cloud jaise platform ko use ho raha hai globally.

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Cloud and other Computing paradigms:

Paradigm	Definition	Cloud relation
Distributed	Multiple systems working on different parts of a task.	Cloud uses distributed computing.
Cluster	connected computers working together like one	Cloud data centers are like clusters.
Grid	large-scale resource sharing (across orgs)	Cloud evolved from grid concept.

6. Essential features of Cloud computing:
1. On demand self service - Bina kisi human interaction ke, resources lena.
2. Broad network access: - Internet ke through access.
3. Resource pooling : Shared resources (multi-tenancy)
4. Rapid Elasticity : Jaldi se scale up/down karna.
5. measured service : usage monitor aur bill hota hai

Characteristic of Cloud Computing
Scalability , Flexibility , Reliability
Security (depends on provider)
Economical
Automatic management.

Advantages of cloud computing

Cost saving
High availability
Disaster recovery
Global accessibility
No infrastructure maintenance.

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Challenges in Cloud computing

- Data security and privacy
- downtime issues
- Limited control over infrastructure
- Compliance and legal issue
- Vendor lock-in.

Ubiquitous cloud and Internet of things (IoT)

Ubiquitous Cloud

Cloud har jagah available hai, chahe mobile ho ya computer isse hi, Ubiquitous computing kehate hain - access anywhere anytime.

IoT connection

IoT devices (smart homes, wearables) ko data store karne aur process karne ke liye cloud ~~of~~ Jarurat hoti hai.

Cloud IoT ko:-

- Real time data processing
- Data analytics
- Storage and scalability deta hai.

Ethical issues in Cloud computing

- Data ownership - data kiska hai?
- Privacy concern - user data misuse na ho
- Transparency - Cloud provider ne kya access kiya?
- Trust - Cloud provider par bharosa karna padta hai
- Vendor lock-in - ek service provider se dusre pe switch mushkil.

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Evaluating the cloud buminers Impact:

benefits

- Cost cutting
- Increased efficiency
- Time saving
- buminers continuity
- Global reach

Evaluation points

- ROI (Return on Investment)
- TCO (Total cost of ownership)
- Risk analysis.

Current trends in Cloud computing.

- Hybrid cloud models
- Serverless computing
- AI and ML on cloud
- Cloud based gaming
- Edge computing.

Future of Cloud computing

- Cloud + Quantum computing
- More AI Integration
- Personal cloud for individuals.
- Sustainable (green) cloud infrastructure.
- 6G+ cloud synergy.

Research Issues in cloud Environment:

- Security and Privacy
- Energy efficient cloud
- Load balancing algorithm
- Cloud resource optimization
- Service level Agreements (SLA's)

Green computing in Cloud

software product matches expected requirement and to ensure that software product is defect free. It involves software/system components using manual and automated tools to evaluate one or more properties of interest. The purpose of software testing is to identify errors, gaps or missing requirement in contrast to actual requirements.

Cloud Computing Architecture and Infrastructure.

1. Cloud Reference model (CRM)

Cloud reference model ek blueprint ya framework hai jo explain karta hai ki cloud system kaise design hota hai, kaunse components involved hote hai aur kaise interact karte hain.

- Ismein major layer hoti hai:

1) Infrastructure layer (IaaS) - Physical and Virtual Resources (compute, storage, network).
2) Platform layer (PaaS) :- Runtime environment, middleware, development tools.
3) Application layer (SaaS) :- End user application (Gmail, google docs etc)

User layer :- Cloud users ya client interface.

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Layers in Cloud Architecture

1. Physical layer
   - Hardware, server, storage, networking, devices.
   - data centers mein hoti hai ye layer
2. Virtualization layer
   - Physical resources ko virtual banata hain
   - Hypervisors (like VMware, KVM) use hote hain
   - Isse multiple VMs (virtual machines) chalti hain ek hi server pe.
3. Infrastructure layer (IaaS):
   - Virtual machines, storage, networks user ko milte hain
   - Example : Amazon EC2, Google compute engine.
4. Platform layer (PaaS)
   - Developers ko environment milta hai apps banane ke liye
   - Example : Google App engine, Heroku etc
5. Application layer
   - End user ko ready made software milta hai internet ke through
   - Example Gmail, microsoft.

Types of Cloud

1. Public Cloud
   - ↳ Open to all, managed by third party provider
   - ↳ Example Aws, Azure, Google Cloud

2. Private cloud,
   - ↳ Specific organization ke liye, secure and customizable
   - ↳ Zyada control or privacy milti hai

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3. Hybrid Cloud
   - Public + Private cloud ka combination
   - sensitive data private mein, baaki public mein

4. Community Cloud
   - Specific group ya community (jaise banks, govt. orgs) ke liye
   - shared resources with similar needs.

Service Models for of Cloud Computing

Model	Definition	User role	Example
IaaS	Infrastructure as a service	User manage OS, apps	AWS EC2, Azure VM
PaaS	Platform as a service	User manage app	Google app engine
SaaS	Software as a service	user just uses the software	Gmail, ms office

Deployment model

1. Public Cloud - hosted by third party, shared infrastructure
2. Private Cloud - hosted privately by one organication
3. Hybrid Cloud - Mix of public + private
4. Community Cloud - Shared among organization with common goals

Data Center design

data center ek physical facility hoti hai jahan cloud infrastructure placed hoti hai, iske kuch major component

Main elements

- Server and storage system
- Cooling system - to prevent overheating
- Power backup - UPS, Generators

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- Security - Biometric access, CCTV
- Fire protection - Smoke detector, extinguishers

Design Goals:
- High availability (24x7 uptime)
- Scalability (easy to expand)
- Energy efficiency
- Redundancy (backup for power, cooling, network)

Interconnection Network
Yeh network backbone hai jo center components ko jodta hai.

Main feature
- High bandwidth - fast data transfer
- Low latency - Quick response time.
- Redundancy - Agar ek link fail ho toh dusra kaam kare
- Virtual LAN (VLANs) - Logical grouping of resources.
- SDN (Software defined networking) - Smart traffic control.

Summary Chart

Section	Key points.
Cloud ref model	Layer: IaaS, PaaS, SaaS
Layers in architecture	physical - Virtualization → IaaS → PaaS → SaaS
Types of cloud	public, private, Hybrid, Community
Service model	IaaS, PaaS, SaaS
Deployment model	Public, private, Hybrid, Community
Data center design	Servers, Cooling, Power security
Interconnection network	High speed, Low latency, SDN